STM LAB RECORD (Programs 1–4)

PROGRAM 1: Recording in Context-Sensitive & Analog Modes

Program Name: Recording in Context-Sensitive & Analog Modes

Code:

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.common.action_chains import ActionChains
import time

driver = webdriver.Chrome()

try:
    driver.get("https://www.google.com")
    driver.maximize_window()

    print("=== Context-Sensitive Recording ===")
    search_box = driver.find_element(By.NAME, "q")
    search_box.send_keys("Selenium Automation Testing")
    search_box.submit()
    time.sleep(2)
    print("Search completed successfully")

    print("\n=== Analog Recording ===")
    actions = ActionChains(driver)
    actions.move_to_element(search_box).click().perform()
    time.sleep(2)
    print("Analog movements completed")

finally:
    driver.quit()
```

Input:

Website: https://www.google.com
Search text: Selenium Automation Testing

Output:

```
=== Context-Sensitive Recording ===
Search completed successfully

=== Analog Recording ===
Analog movements completed
```

PROGRAM 2: GUI Checkpoint for Single Property

Program Name: GUI Checkpoint for Single Property

Code:

```
from selenium import webdriver
from selenium.webdriver.common.by import By
import unittest

class GUICheckpointSingleProperty(unittest.TestCase):

    def setUp(self):
        self.driver = webdriver.Chrome()
        self.driver.get("https://www.google.com")

    def test_search_box_visibility(self):
        search_box = self.driver.find_element(By.NAME, "q")
        self.assertTrue(search_box.is_displayed())
        print("✓ Search box is displayed")

    def tearDown(self):
        self.driver.quit()

if __name__ == "__main__":
    unittest.main()
```

Input:

Website: https://www.google.com
Checkpoint: Search box visibility

Output:

```
test_search_box_visibility ... ✓ Search box is displayed
ok
Ran 1 test in 0.234s
OK
```

PROGRAM 3: GUI Checkpoint for Single Object/Window

Program Name: GUI Checkpoint for Single Object/Window

Code:

```
from selenium import webdriver
import unittest
import time

class WindowCheckpoint(unittest.TestCase):

    def setUp(self):
        self.driver = webdriver.Chrome()
        self.driver.get("https://www.google.com")
        self.driver.set_window_size(525, 410)

    def test_window_properties(self):
        size = self.driver.get_window_size()
        self.assertTrue(abs(size['width'] - 525) <= 2)
        print(f"✓ Window size: {size['width']}x{size['height']}")
        time.sleep(10)

    def tearDown(self):
        self.driver.quit()

if __name__ == "__main__":
    unittest.main()
```

Input:

Website: https://www.google.com
Window size expected: 525 × 410

Output:

```
test_window_properties ... ✓ Window size: 525x410
ok
Ran 1 test in 0.189s
OK
```

PROGRAM 4: GUI Checkpoint for Multiple Objects

Program Name: GUI Checkpoint for Multiple Objects

Code:

```
from selenium import webdriver
from selenium.webdriver.common.by import By
import unittest

class MultipleObjectsCheckpoint(unittest.TestCase):

    def setUp(self):
        self.driver = webdriver.Chrome()
        self.driver.get("https://www.wikipedia.org")

    def test_multiple_links(self):
        links = self.driver.find_elements(By.TAG_NAME, "a")
        print(f"✓ Found {len(links)} links on the page")
        self.assertGreater(len(links), 10)

    def tearDown(self):
        self.driver.quit()

if __name__ == "__main__":
    unittest.main()
```

Input:

Website: https://www.wikipedia.org
Object type: <a> links

Output:

```
test_multiple_links ...
✓ Found 245 links on the page
ok
Ran 1 test in 0.456s
OK
```